Category Theory for Mathematical Physics — Exercise Sheet 6

F-symbols and the pentagon

Variant: Questions only

Conventions as in Lecture 6: the categories are multiplicity-free, with vertices ψ_c^{ab}; the F-matrix F_d^{abc} has rows e and columns f in the order of the list of simple objects (Definition 6.1); an F-symbol with a unit leg is 1 when its trees exist, and every symbol whose trees do not exist is 0. Fibonacci has $1, \tau$ and Ising $1, \sigma, \psi$ (the bare ψ is Ising's fermion; a vertex always carries its labels). $\mathbb{Z}_2 = \{1, s\}$ is written multiplicatively, as in Lecture 3. The gauge numbers are u_c^{ab} (Theorem 6.4); $\varphi = (1+\sqrt{5})/2$; H is the Hadamard matrix; x, A, B, C, D are the Fibonacci unknowns of Theorem 6.6. In Exercise 5, $\uparrow, \downarrow$ are the spin-$\frac{1}{2}$ states $m = \pm\frac{1}{2}$. In Exercise 6, $\gamma_1, \gamma_2, \gamma_3$ name three F-symbols (the labels g of the pentagon and the group elements g of Lecture 3 are other letters).

Exercise 1 (Boson and semion). Take the fusion ring of $\mathbb{Z}_2$: $s \otimes s \cong 1$.

(i) Show that the only F-symbol without a unit leg is $x = [F_s^{sss}]_{11}$, and that the pentagon equation of Theorem 6.3 reduces to $x^2 = 1$.

(ii) Show that x is gauge invariant, and conclude that the fusion ring has exactly two solutions up to gauge. Which categories of Lecture 3 are they, and what is κ_s for each?

(iii) Draw the F-move Equation (6.2) for $g = h = k = s$.

Exercise 2 (Fibonacci in terms of d_τ, and Yang–Lee). Let d_τ be a root of $d^2 = 1 + d$, and let $d_\tau^{-1/2}$ be a square root of d_τ^{-1}.

(i) Show that $x = 1$ and

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} d_\tau^{-1} & d_\tau^{-1/2} \\ d_\tau^{-1/2} & -d_\tau^{-1} \end{pmatrix}$$

satisfy the six equations listed before Theorem 6.6, and that $d_\tau = \varphi$ gives the unitary solution there.

(ii) For $d_\tau = \varphi$, write the inverse F-move Equation (6.3) for three τ's with total charge τ whose last two fuse to 1.

(iii) For $d_\tau = -1/\varphi$ (Yang–Lee) write the matrix, and show that it squares to the identity but is not unitary. Compute $\kappa_\tau = d_\tau A$ and $1 + d_\tau^2$ for both roots, and compare the second with $\dim \mathrm{Fib} = 2 + \varphi$ under $\sqrt{5} \mapsto -\sqrt{5}$. Why is Yang–Lee no anyon model?

Exercise 3 (What a gauge can change). (i) For Fibonacci, with $u_1^{\tau\tau} = p$ and $u_\tau^{\tau\tau} = q$, compute from Theorem 6.4 the factors on x, A, B, C and D. Show that BC is invariant and B/C is multiplied by q^4/p^2. The numbers $x = 1$, $A = \varphi^{-1}$, $B = 1$, $C = \varphi^{-1}$, $D = -\varphi^{-1}$ solve the pentagon; can a gauge with $|u| = 1$ make them unitary?

(ii) For Ising, show that $[F_\sigma^{\sigma\sigma\sigma}]_{11}$ is invariant, and compute $F_\sigma^{\sigma\sigma\sigma}$ and $[F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma}$ after the gauge $u_\psi^{\sigma\sigma} = i$, all other u equal to 1. Is the new matrix unitary?

Exercise 4 (Pentagon instances by hand). (i) Write out the pentagon equation of Theorem 6.3 for Fibonacci with $a = b = c = d = e = g = k = \tau$ and $f = l = 1$, and show that it is $A^2 + BCD = 0$. Draw the five trees of the pentagon for these labels.

(ii) Find labels whose instance is $BD = xBD$.

(iii) For Ising, with $a = b = c = d = g = k = \sigma$ and $e = f = l = 1$, show that the instance reads

$$1 = \tfrac{\kappa^2}{2}\big(1 + [F_1^{\sigma\psi\sigma}]_{\sigma\sigma}\big)$$

when $F_\sigma^{\sigma\sigma\sigma} = \kappa H$, and check it for the data of Theorem 6.7.

Exercise 5 (SU(2) recoupling is a $6j$ symbol). In $\mathrm{Rep}(\mathrm{SU}(2))$ (semisimple, rigid and multiplicity-free, though not finite) the F-symbols are defined as in Definition 6.1. Take isometric vertices given by the Clebsch–Gordan coefficients:

$$\psi_0^{\frac12\frac12}(1) = \tfrac{1}{\sqrt2}(\uparrow\downarrow - \downarrow\uparrow),$$
$$\psi_1^{\frac12\frac12}\colon\ |1,1\rangle \mapsto \uparrow\uparrow,\ |1,0\rangle \mapsto \tfrac{1}{\sqrt2}(\uparrow\downarrow + \downarrow\uparrow),$$
$$\psi_{\frac12}^{1\frac12}\colon\ \uparrow \mapsto \sqrt{\tfrac23}\,|1,1\rangle\downarrow - \sqrt{\tfrac13}\,|1,0\rangle\uparrow,$$
$$\psi_{\frac12}^{\frac121}\colon\ \uparrow \mapsto \sqrt{\tfrac13}\uparrow|1,0\rangle - \sqrt{\tfrac23}\downarrow|1,1\rangle,$$

and $\psi_{\frac12}^{0\frac12} = \psi_{\frac12}^{\frac120} = \mathrm{id}$.

(i) Explain why rescaling the vertices to the normalisation Equation (5.1) does not change the F-symbols. Compute the trees L_0, L_1, R_0, R_1 for three spin-$\frac12$ with total spin $\frac12$ on the state $\uparrow$, and the matrix $F_{\frac12}^{\frac12\frac12\frac12}$.

(ii) Compare with the Racah–Wigner formula

$$[F_d^{abc}]_{ef} = (-1)^{a+b+c+d}\sqrt{(2e+1)(2f+1)}\begin{Bmatrix} a & b & e \\ c & d & f\end{Bmatrix},$$

given that the $6j$ symbols $\left\{\begin{smallmatrix} 1/2 & 1/2 & e \\ 1/2 & 1/2 & f\end{smallmatrix}\right\}$ are $-\frac12, \frac12, \frac12, \frac16$ for $(e,f) = (0,0), (0,1), (1,0), (1,1)$. Compute $d_{1/2}[F_{\frac12}^{\frac12\frac12\frac12}]_{00}$ with $d_{1/2} = 2$.

(iii) Write the pentagon instances with $a = b = c = d = \frac12$, $e = 0$, $g = k = \frac12$ (one for each f, l). Given that the 1×1 symbols in them are 1, show that together they say $(F_{\frac12}^{\frac12\frac12\frac12})^2 = 1$. (This instance is degenerate: (i) already implies it. The general pentagon for these F-symbols is the Biedenharn–Elliott identity, which the check script verifies on representations built with rep-engine.)

Exercise 6 ⋆. *The Ising F-symbols, and why there are no others.* Name the Ising F-symbols without a unit leg as follows:

$$n_1 = [F_\psi^{\psi\psi\psi}]_{11},\ n_2 = [F_\sigma^{\psi\psi\sigma}]_{1\sigma},$$
$$n_3 = [F_1^{\psi\sigma\sigma}]_{\sigma\psi},\ n_4 = [F_\psi^{\psi\sigma\sigma}]_{\sigma 1},$$
$$n_5 = [F_\sigma^{\psi\sigma\psi}]_{\sigma\sigma},\ n_6 = [F_\psi^{\sigma\psi\sigma}]_{\sigma\sigma},\ n_7 = [F_1^{\sigma\sigma\psi}]_{\psi\sigma},$$
$$\gamma_1 = [F_1^{\sigma\psi\sigma}]_{\sigma\sigma},\ \gamma_2 = [F_\sigma^{\sigma\psi\psi}]_{\sigma 1},$$
$$\gamma_3 = [F_\psi^{\sigma\sigma\psi}]_{1\sigma},$$

and $F_\sigma^{\sigma\sigma\sigma} = (m_{ef})$ with $e, f \in \{1, \psi\}$.

(i) Compute the gauge factors of $\gamma_1, \gamma_2, \gamma_3$, and show that every solution of the pentagon with invertible F-matrices (as for any category) is gauge equivalent to one with $\gamma_1 = \gamma_2 = \gamma_3 = 1$.

(ii) With $\gamma_1 = \gamma_2 = \gamma_3 = 1$ the following are pentagon instances:

$$n_1 = n_7, \quad n_1 n_2 = n_2, \quad n_2 = 1,$$
$$n_6^2 = n_2, \quad n_2 n_5^2 = n_2, \quad n_1 = n_2 n_3 n_4, \quad n_3 n_7 = n_4,$$
$$m_{1\psi} = m_{11}, \quad m_{\psi 1} = n_4 m_{11},$$
$$m_{1\psi} m_{\psi 1} + m_{11}^2 = 1, \quad n_6 m_{1\psi} m_{\psi 1} + m_{11}^2 = 0,$$
$$n_5 m_{\psi\psi} = n_4 m_{1\psi}, \quad n_5 m_{\psi 1} = n_7 m_{\psi\psi},$$
$$m_{1\psi} m_{\psi\psi} + m_{11} m_{1\psi} = 0.$$

Derive the last one from the labels $a = b = c = d = g = k = \sigma$, $e = f = 1$, $l = \psi$. Then show that these fourteen equations have exactly two solutions: $F_\sigma^{\sigma\sigma\sigma} = \kappa H$ with $\kappa = \pm 1$, $n_5 = n_6 = -1$, and every other symbol 1.

(iii) Conclude that the Ising fusion ring carries exactly two fusion categories up to gauge, and that they are told apart by κ.